

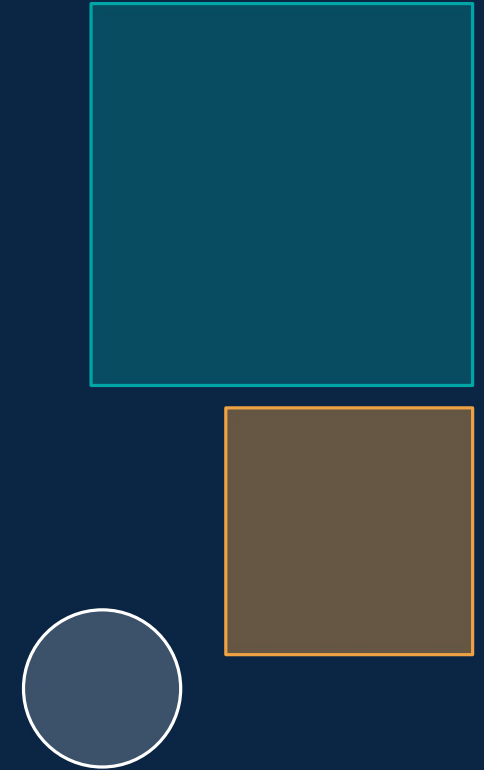
THESIS PROJECT · POLITECNICO DI TORINO

AROL Customer Platform

Agentic AI for Fleet Management



Development of the chatbot, the AI Agents architecture and the orchestrator within AROL's Digital Business Transformation journey



WHO WE ARE

AROL at a glance

AROL Group is a global benchmark in the design and manufacturing of capping machines and primary packaging equipment, serving the beverage, food, personal care, home care and pharma industries.

Capping & Closing

Filling

Inspection

IoT & Smart Factory

Headquarters: Canelli (AT) · Commercial and service presence across Europe, the Americas and Asia.

40+

Years of expertise in capping

Global

Service & customer base

Industry 4.0

Connected, IoT-ready machines

R&D

Continuous product innovation

The context: Digital Business Transformation

A strategic journey to bring AROL and its customers into the smart factory era: data, processes and products integrated into a single digital ecosystem.

01

Connectivity

Machines, processes and people connected: from CRM to ERP, from IoT to service.

02

Data as an asset

Collection, normalization and monetization of all data linked to the machines.

03

Artificial Intelligence

Generative and agentic AI supporting customers, service teams and decision making.

The Customer Platform

A single digital access point

A portal dedicated to end customers that enables direct, immediate communication between AROL and the users of its machines.

It becomes the digital archive of the customer's know-how, with structured access to all the information about their fleet of machines.

"From data to knowledge."

1

Advanced self-service

24/7 access to machine data and documents.

2

End-to-end transparency

From quote to after-sales on a single timeline.

3

AI-driven support

Chatbot and AI agents for troubleshooting and operations.

4

Complete fleet management

Consolidated view of the entire customer machine park.

What the platform collects

01

Quotes

Initial quote and the complete history of revisions.

02

Orders

Status and details of every customer order.

03

Contractual documents

Contracts, attachments and clauses always available.

04

Use & maintenance manuals

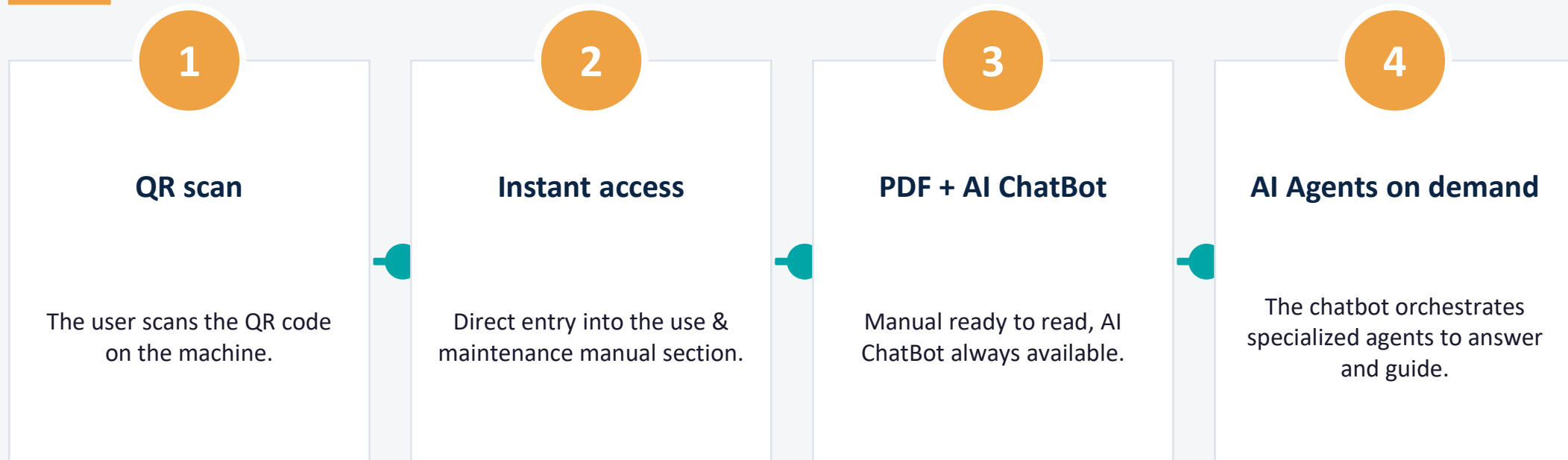
Up-to-date and searchable technical documentation.

05

IoT data

Real-time and historical telemetry from the machines.

Use case: from QR code to AI support



Goal: shorten the time to access information and actively support troubleshooting in the field.

From digital archive to Agentic AI

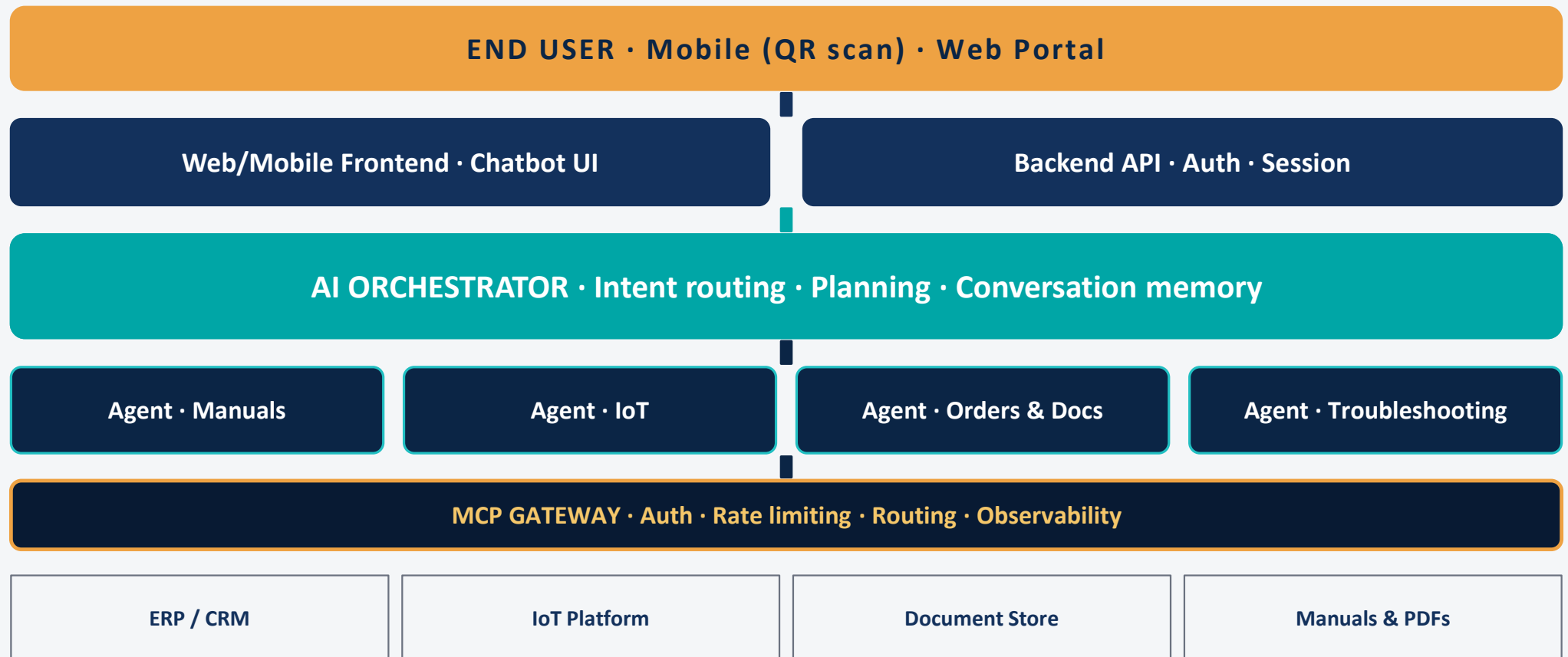
TODAY · Digital archive

- Documents, manuals and IoT centralized in a single portal
- Keyword search and manual navigation
- Knowledge is available but not immediately actionable
- The user searches: information is static

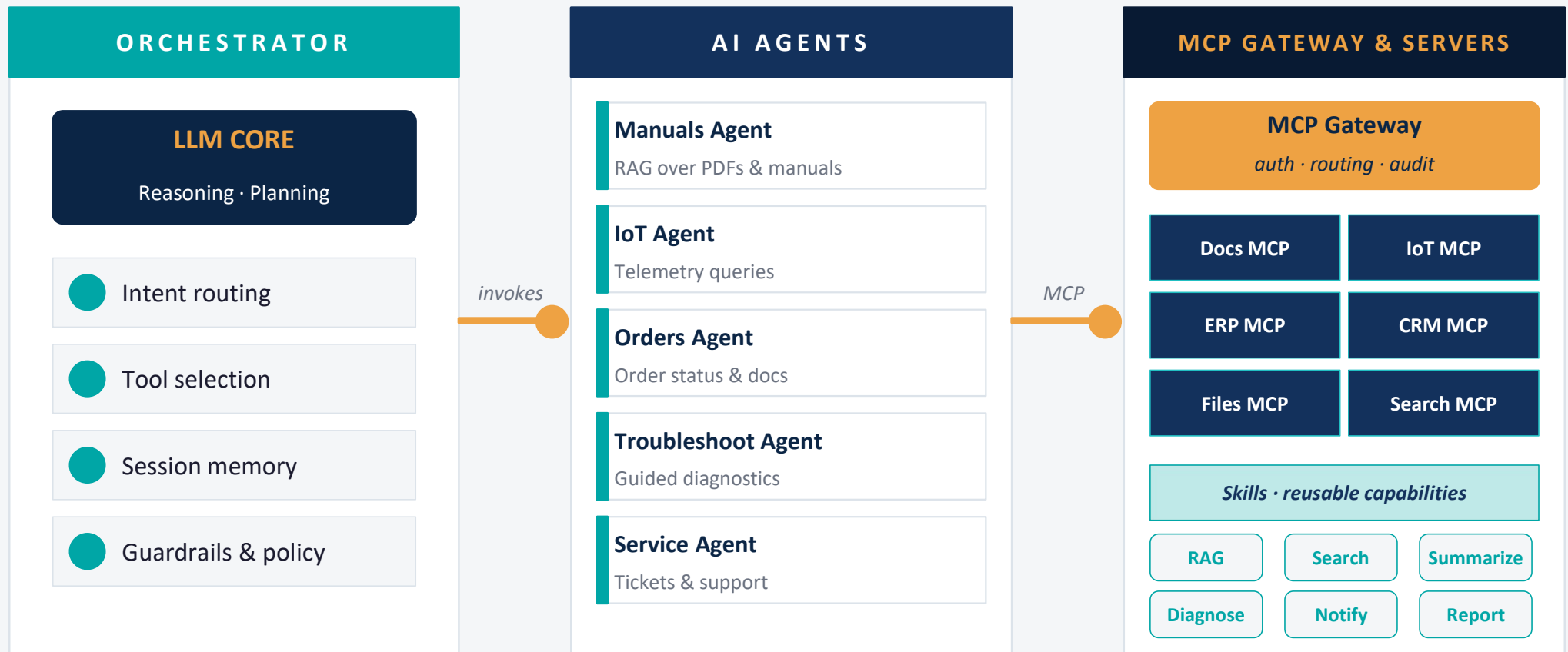
TOMORROW · Agentic AI

- AI searches, synthesizes and proposes autonomously
- Specialized agents for manuals, IoT, orders, service
- Guided troubleshooting on real machine scenarios
- Information comes to you: knowledge becomes actionable

Logical system architecture



AI Architecture: MCP, Gateway, Skills



Tech stack & DevOps

Frontend	Backend & API	AI Layer	DevOps & Platform
• React / Next.js • TypeScript • Custom components • Mobile / scalable	• Python (FastAPI) or	• LLM (Azure OpenAI)	• GitHub / GitLab +

Feel free to use your preferred stack
The focus is the architectural design

Scope of work for the students

01 Chatbot Back-end

APIs, session management, integration with LLM and Orchestrator. Conversation persistence, authentication, logging.

REST/WS API

Auth

Session store

02 Chatbot Front-end

Conversational UI embedded in the portal, mobile-first for QR access. Streaming responses, citations, attachments.

React/Next.js

Streaming UI

Mobile UX

03 AI Agents Architecture

Design and implementation of the specialized agents (Manuals, IoT, Orders, Troubleshooting) with data access via MCP.

Agents

Tools

MCP servers

04 Orchestrator

Intent routing, agent selection and coordination, context management and safety guardrails.

Routing

Planning

Guardrails

Thank you



Questions & discussion

*Ready to build together the AROL customer platform
and the next generation of AI-powered experiences.*

AROL Group · IT Department · Canelli (AT)



Your contact points

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